MedBridge: A Policy-Grounded, Human-in-the-Loop LLM Agent and Chatbot for Healthcare Prior Authorization

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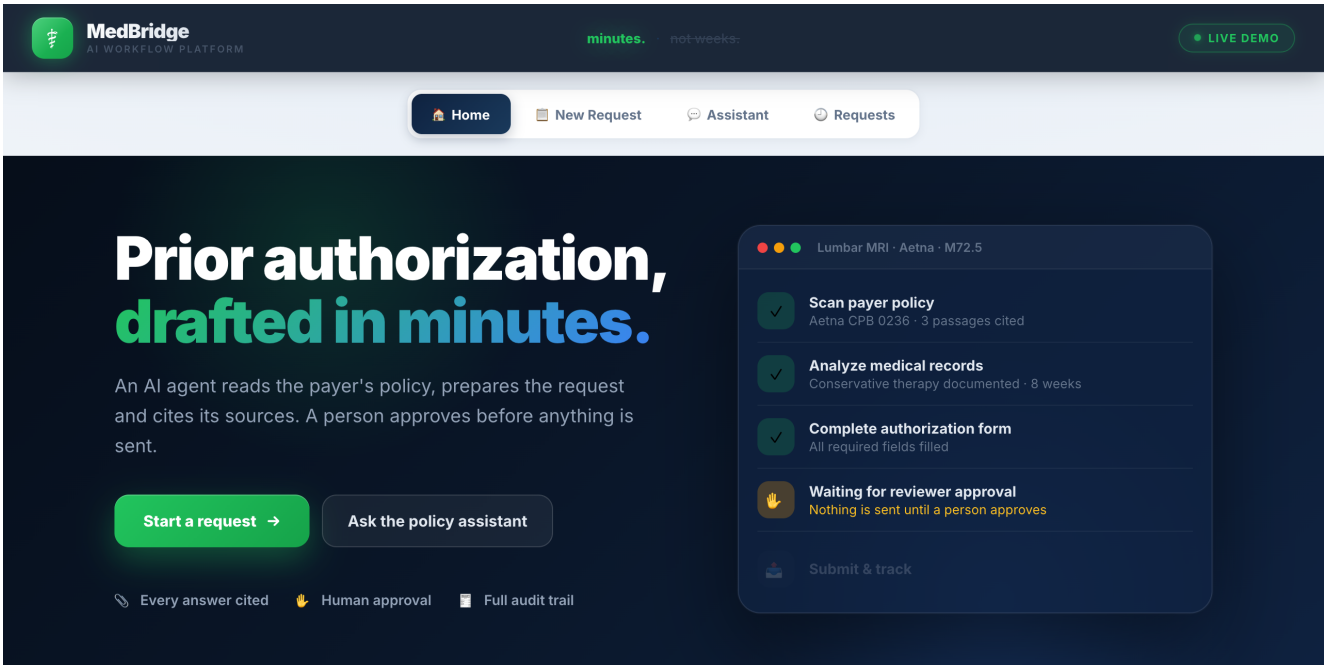


Figure 1: MedBridge prototype home screen. The animated card previews the agent's workflow: policy scan, record analysis, form completion, a pause for human approval, then submission and tracking. Screenshot of our own prototype (synthetic demo case). The project began as the 1st-place solo hackathon entry described in Section 2.4.

1. Abstract

Prior authorization (PA) makes clinics obtain insurer approval before many tests and treatments, a largely manual process that delays care; denials are rarely appealed although most appeals succeed. We extend *MedBridge*, a 1st-place solo hackathon prototype, into a domain-specific assistant that takes one structured request (patient, payer, service, diagnosis, justification) and uses retrieval-augmented generation (RAG) over public payer and CMS policies plus a tool-using LLM agent to prepare the form, pause for human approval, submit, track the decision and, on denial, draft an appeal citing policy text. A companion chatbot answers coverage questions with sources. Outputs are an audited request or cited appeal draft for clinic staff. Payer and patient-record interfaces are simulated.

2. Introduction

2.1. Target Market

Prior authorization is a documented burden. In the American Medical Association's 2024 survey of 1,000 physicians, 94% said it delays necessary care, 24% reported a serious adverse event for a patient, and practices reported completing 43 requests and spending about 12 staff hours per physician each week [1] (the AMA opposes prior authorization, so these are advocacy-group survey results). Denials are common and mostly reversed when challenged: in 2024 Medicare Advantage insurers fully or partially de-

nied 4.1 million of 52.8 million requests, yet 80.7% of appealed denials were overturned while only 11.5% of denials were appealed [7]. Reported waits are contested: about two-thirds of physicians in an earlier AMA survey (2017 data) waited at least one business day and roughly 30% three days or longer [2], whereas insurer-reported median response times are about one day [8]. Manual handling is costly: in the 2019 CAQH Index a manual prior authorization cost providers $10.92 against $1.88 electronically [4]. Federal rule CMS-0057-F additionally requires large payers (Medicare Advantage, Medicaid managed care and exchange plans) to offer prior-authorization APIs by January 2027 [5], so tooling that speaks structured requests is becoming relevant.

2.2. Target Users

The primary users are prior-authorization coordinators and administrative staff in outpatient clinics, who assemble requests from payer rules and chart notes. The secondary users are physicians, who review and sign appeals. Their needs are (i) knowing what a specific payer requires for a specific service, (ii) a correct, complete request the first time, (iii) a defensible appeal when a request is denied, and (iv) a record of who approved what.

2.3. Why General LLMs Fall Short

A general-purpose chatbot lacks current payer-specific criteria, cannot act on systems (prepare, submit, track, appeal), and gives untraceable answers, which is unacceptable when a wrong coverage claim ends up in a submission. MedBridge addresses these gaps with two techniques from the course: **Knowledge Augmentation & Retrieval** (policy documents embedded in a vector database and retrieved per request) and **Agentic Workflows & Tool Use** (a reasoning loop that chooses among tools and branches on denial).

2.4. Project Origin: Hackathon Winner, Then Extended

MedBridge won 1st place at the AI Agents Hackathon (OSS4AI × Beta Fund × Gravitational Ventures), built solo in 36 hours [9]. This paper describes how that prototype is rebuilt and extended for this course.

The hackathon version was a working agent: an LLM-driven loop over six tools (policy scan, record analysis, form fill, submission, status tracking, appeal drafting), a React interface that shows each step live, and simulated payer responses. Its policy data was hand-written, it submitted without a review step, and it had no evaluation. The extension for this course adds:

- retrieval over real public payer and CMS policy documents, replacing hand-written policy data;
- a human approval gate and a persistent audit trail;
- a cited policy chatbot;
- a retrieval evaluation and a planned agent evaluation;
- a provider-independent runner that adds AWS Bedrock.

3. Related Work

Retrieval-augmented generation. Lewis *et al*. [10] combine a parametric generator with a non-parametric retriever to ground outputs in external text. Xiong *et al*. [18] benchmark RAG for medical question answering and report gains over prompting without retrieval. We apply the same principle to payer coverage policy, where documents differ by payer and change over time.

Tool-using agents. ReAct [19] interleaves reasoning with actions and observations, and Toolformer [14] trains models to call APIs. MedBridge uses native function calling in a ReAct-style loop in which the model, not a fixed script, selects the next tool.

Clinical language models. Singhal *et al*. [16] show that LLMs encode clinical knowledge but still need expert evaluation. We avoid relying on parametric medical knowledge and retrieve criteria at inference time.

LLMs for prior authorization. Pandey *et al*. [11] split medical-necessity justification across specialized LLM agents and report high checklist accuracy with GPT-4. Vatsal *et al*. [17] frame prior authorization as guideline-based question answering over patient records and report a mean weighted F1 of 0.61. Awan and Raza [3] find that LLM-drafted prior-authorization letters have strong clinical content but weak administrative elements such as missing billing codes. Selvaraj *et al*. [15] study the fairness of an automated prior-authorization system.

Industry. Vendors such as Cohere Health offer AI-assisted prior-authorization platforms to health plans and providers [6], and patient-facing tools draft denial appeals [12]; the success rates reported for these products are company claims.

Gap. The research works above evaluate judgments or letters in isolation, and the products serve payers or patients. To our knowledge, none combines retrieval over real payer policy documents, a tool-using agent, a human approval gate and denial-triggered appeal drafting with citations for clinic staff, which is the combination explored here. (The literature search will be extended for Phase 2.)

4. Technical Contribution

Novelty. The hackathon prototype supplied the agent loop and interface; the contributions below are new in this work. The contribution is an end-to-end, auditable pipeline rather than a new model: policy-grounded generation, agentic action, and enforced human review in one clinic-side workflow, with an evaluation protocol for each stage.

1. **Policy-grounded retrieval with payer scoping.** Public policy documents (Aetna Clinical Policy Bulletins, CMS National Coverage Determination 220.2, Blue Cross Blue Shield, UnitedHealthcare and Cigna policies) are chunked (160 words, 30-word overlap), embedded with all-MiniLM-L6-v2 [13] and stored in ChromaDB with source URL, payer and retrieval date. Queries are filtered to the requesting payer, and a payer with no documents returns nothing rather than another payer's policy.

2. **Human-in-the-loop gate inside the agent loop.** Outward-facing tools are marked as requiring approval. The loop blocks before such a tool runs until a reviewer approves or rejects, and every action is written to a persistent audit trail.

3. **Denial-conditioned appeal drafting with citation.** After a denial, the agent retrieves the policy passage that states the relevant criterion (for example, the conservative-therapy requirement) and quotes it in the draft appeal, which still requires physician review.

4. **Cited policy chatbot.** A conversational endpoint answers coverage questions only from retrieved passages, shows each source, and states when the collection does not cover a payer.

5. **Vertical-plugin architecture.** Tools, prompts and form schema live in one module per vertical; the agent runner, API and UI are generic. The runner supports Groq, Claude and AWS Bedrock models.

5. Preliminary System Design

Figure 2 shows the components. A request flows: UI → FastAPI (background task) → agent runner. The runner sends the request and tool schemas to the LLM, executes whichever tool the model selects, appends the observation, and repeats until the model stops calling tools. The six tools are `scan_insurance_policy` (RAG), `analyze_medical_records`, `fill_authorization_form`, `submit_authorization`, `track_authorization_status` and `draft_appeal`. Before `submit_authorization` runs, the gate sets the request to *awaiting approval*; approval resumes the loop and rejection ends it. The chatbot shares the retriever but uses its own tool loop.

Inputs and outputs. Input: payer, member and patient identifiers, service, ICD-10 code and clinical justification, or a free-text policy question. Output: a completed and reviewed request with a step log and audit trail, a cited appeal draft on denial, or a cited answer.

Data. The policy index is built from publicly available documents downloaded from their publishers (Aetna, CMS, Blue Cross Blue Shield of South Carolina, UnitedHealthcare, Cigna), with the retrieval date recorded per file. Patient records, payer decisions and submission are *simulated* in the prototype, and no real patient data is used.

Prototype status. A working prototype implements the pipeline above. As a first check of retrieval, we labeled 29 questions with the document that answers each; with payer filtering, Recall@3 is 1.00 and MRR is 0.92, against Recall@3 of 0.86 and MRR of 0.70 without it. The questions were written by the author after reading the documents and the corpus is small (10 documents), so this is a sanity check, not a benchmark.

Planned evaluation (Phase 2). (1) Retrieval quality with chunk-size and filter ablations; (2) agent tool-sequence correctness over repeated runs on several LLM providers; (3) citation faithfulness of drafted appeals and chat answers, judged on a sample; (4) agent with RAG versus a static policy table.

Limitations of the current design. Simulated payer and EHR interfaces, no user authentication or roles, no HIPAA-grade handling of real patient data, a small policy corpus without Humana or Medicaid documents, and drafts that always require clinician review.

6. Plan for Phase 2

- By Nov 10: run the evaluation of the agent on a hosted LLM provider; fix tool-order failures.
- By Nov 17: replace hard-coded patient records with generated synthetic patients stored in the database.
- By Nov 24: extend the policy corpus; build the chatbot evaluation set.
- By Dec 1: ablations (retrieval settings, with versus without RAG).
- By Dec 8: complete methodology, results, discussion and limitations.
- Dec 15: final paper and code submission.

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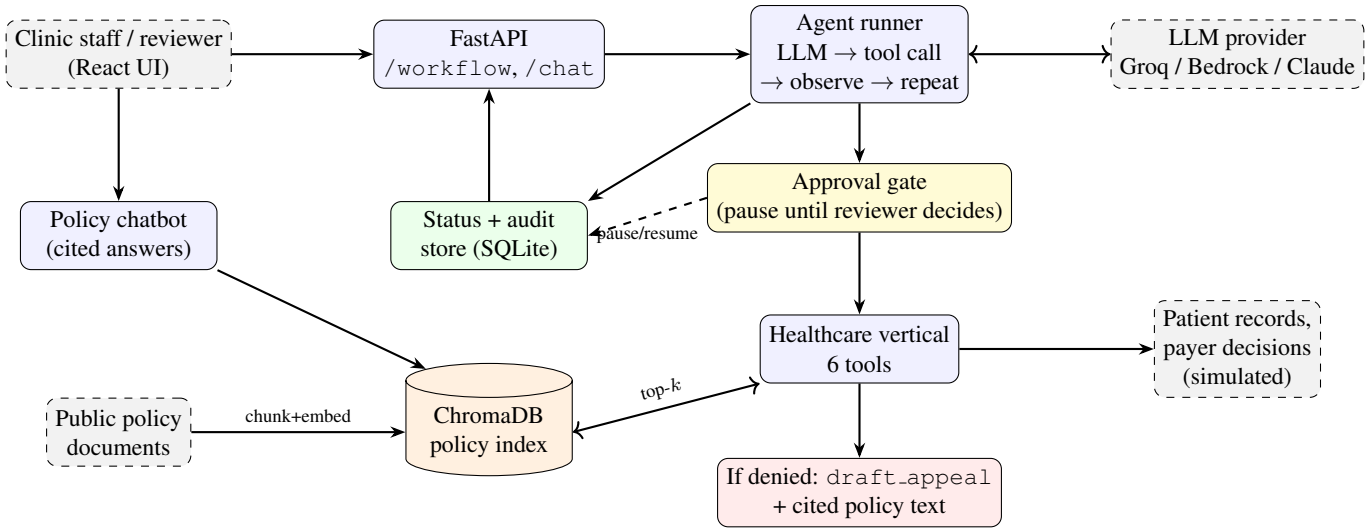


Figure 2: Preliminary system design. Staff use the chatbot or submit a structured request. The agent loop calls tools, retrieving payer policy passages from the vector index; before the submission tool runs, the approval gate pauses the run until a reviewer decides. Status and the audit trail are persisted. A denial triggers a cited appeal draft. Dashed boxes are external to our code; patient records and payer submission are simulated.

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